

Francesco Beretta
(CNRS / University of Bern / University of Neuchâtel)

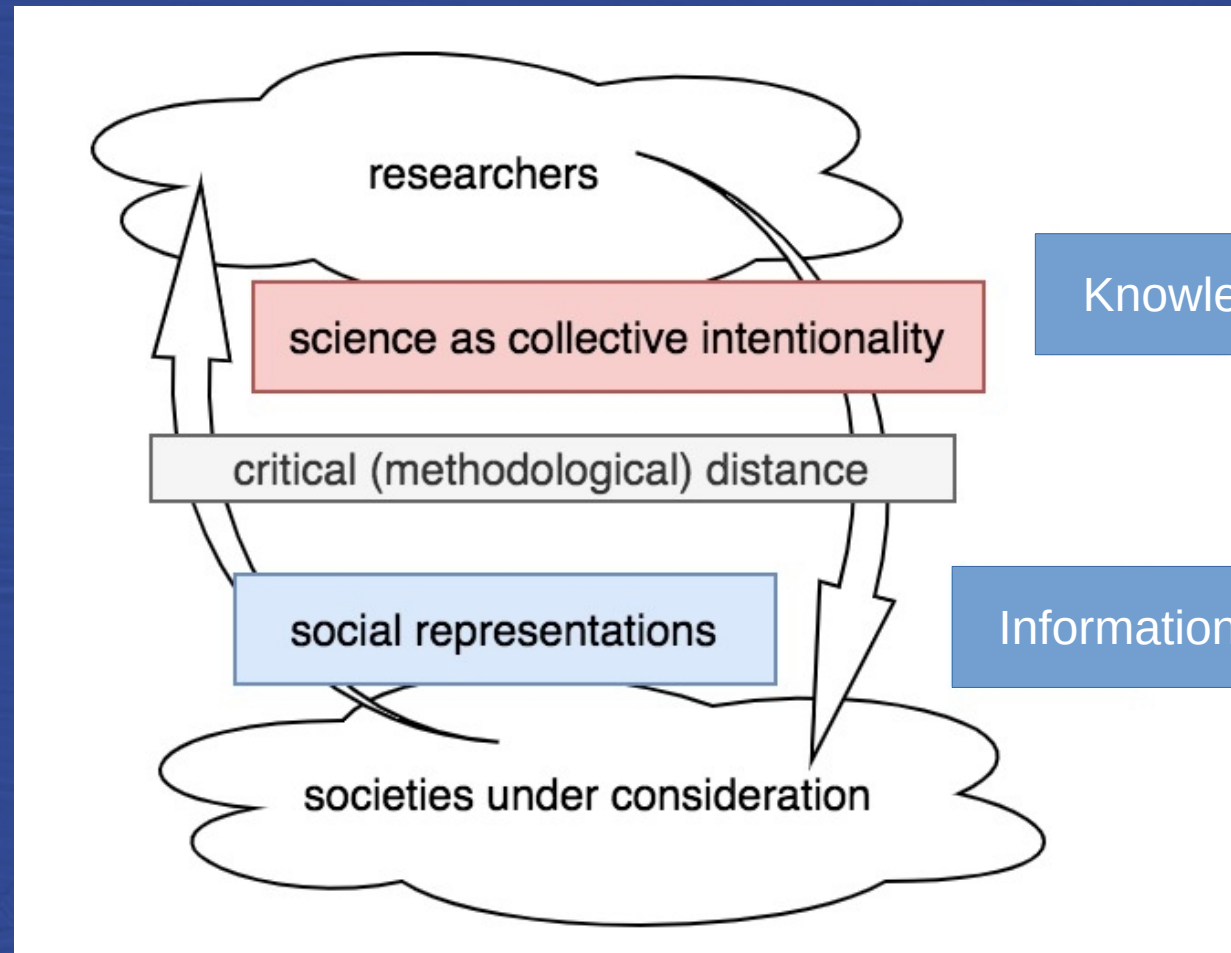
Conceptual Modelling and the Cycle of Knowledge Production in the HSS

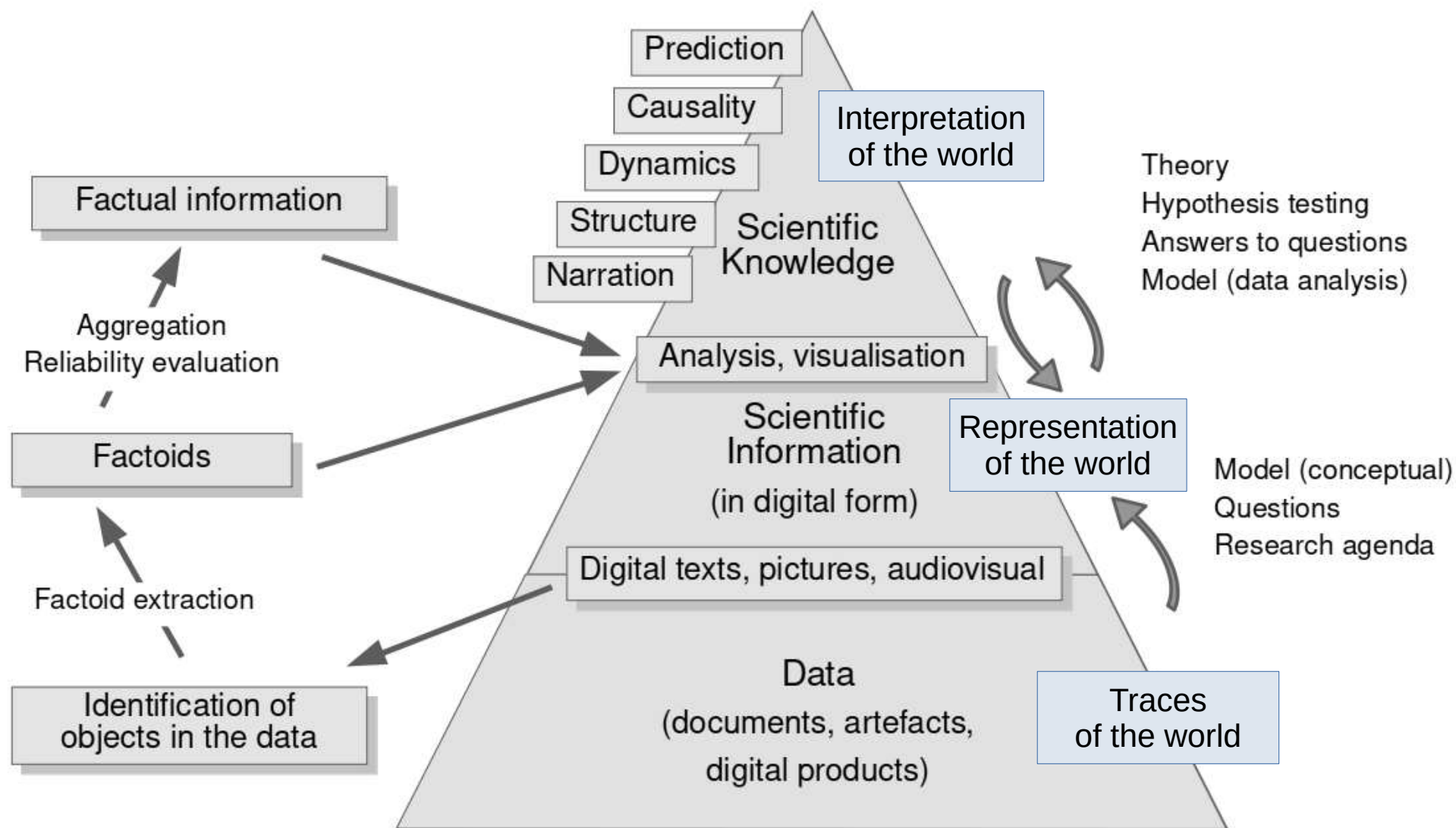
University of Bern
(version 1.0, october 2026)

Still ongoing – not finished !



Scientific knowledge as a portion of social representations





Information as representation of the world :

**Social
Representations
(Collective
Intentionality)**



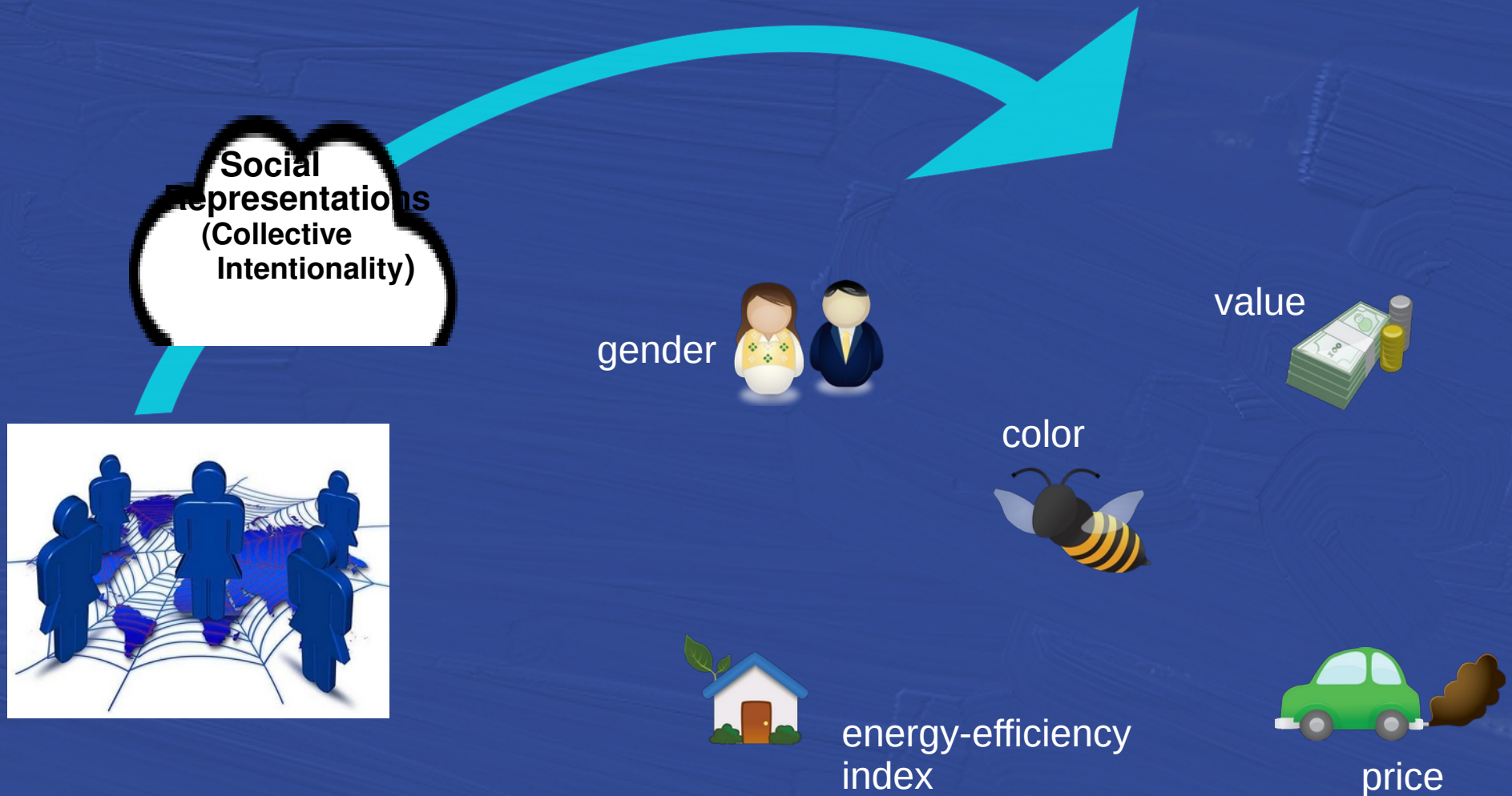
Information as **representation** of the world :

- representation of the **objects** in the world



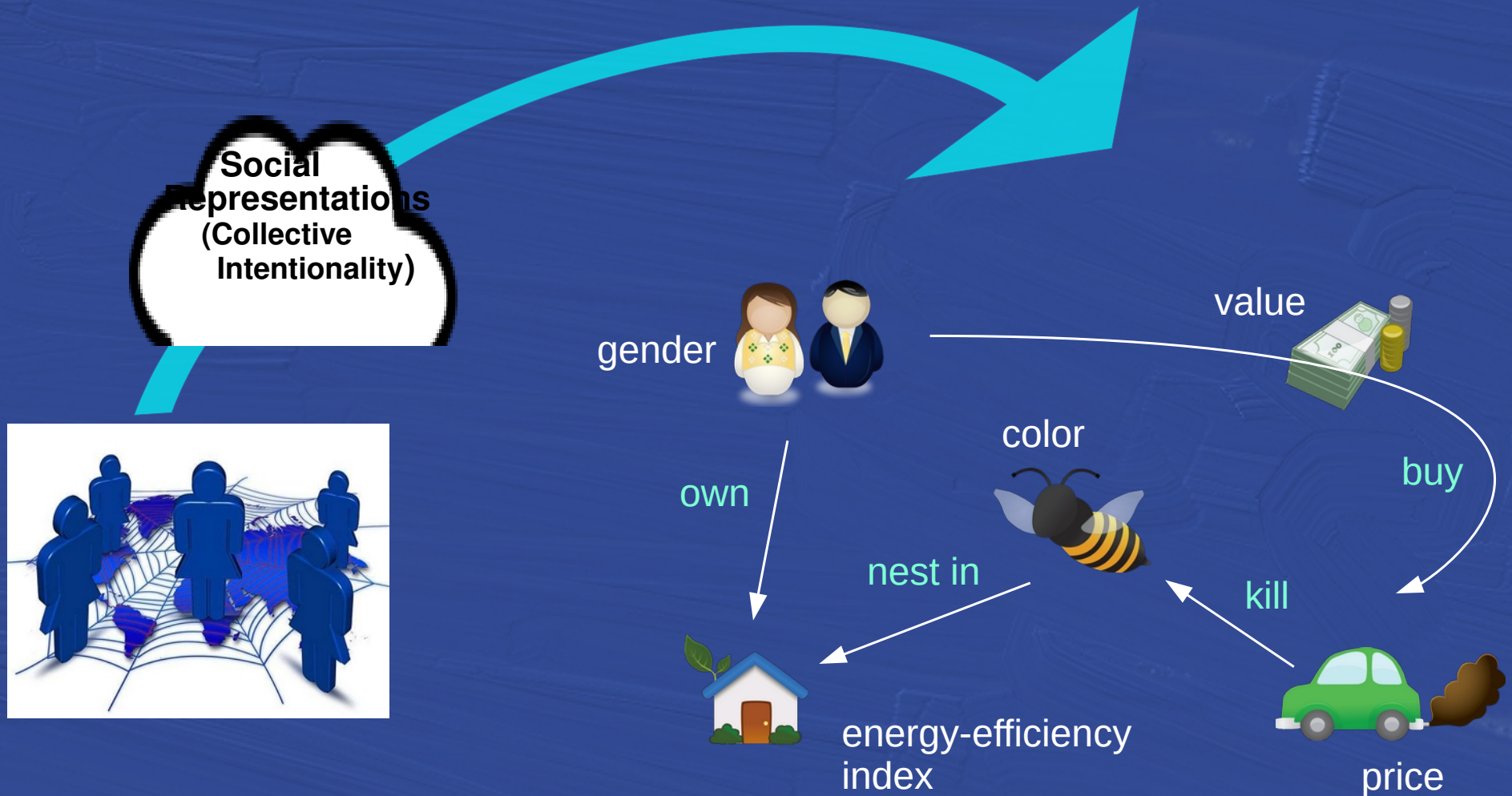
Information as **representation** of the world :

- representation of the **objects** in the world
- of their **properties** (qualities)



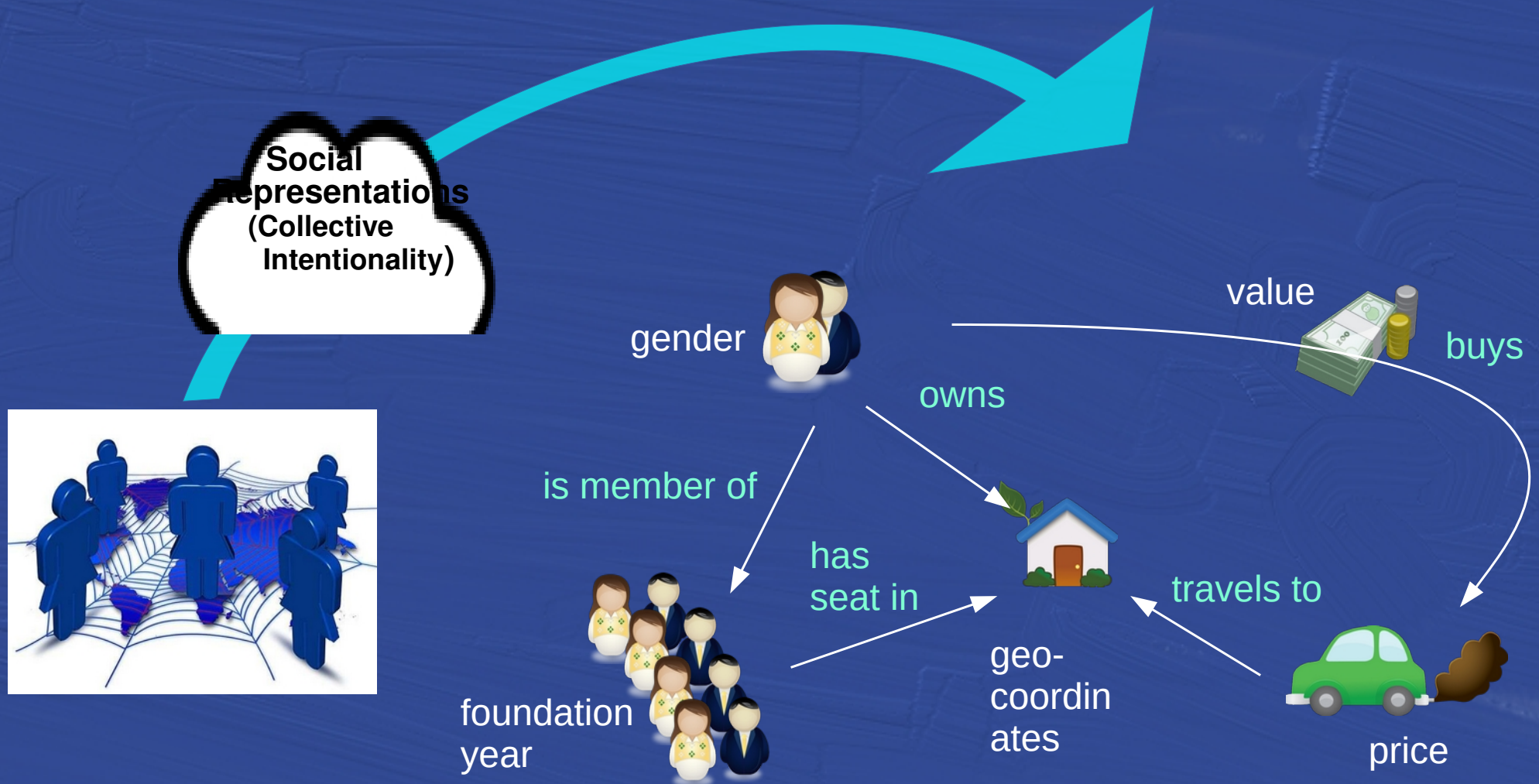
Information as **representation** of the world :

- representation of the **objects** in the world
- of their **properties** (qualities)
- of their **relationships**



Information as **representation** of the world :

- representation of the **objects** in the world
- of their **properties** (qualities)
- of their **relationships**



« In 1592, he [Galileo Galilei] moved to the University of Padua where he taught geometry, mechanics, and astronomy until 1610 ».

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Letter by cardinal Bellarmine to Paolo Antonio Foscarini, 12 April 1615:

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Expression of an opinion

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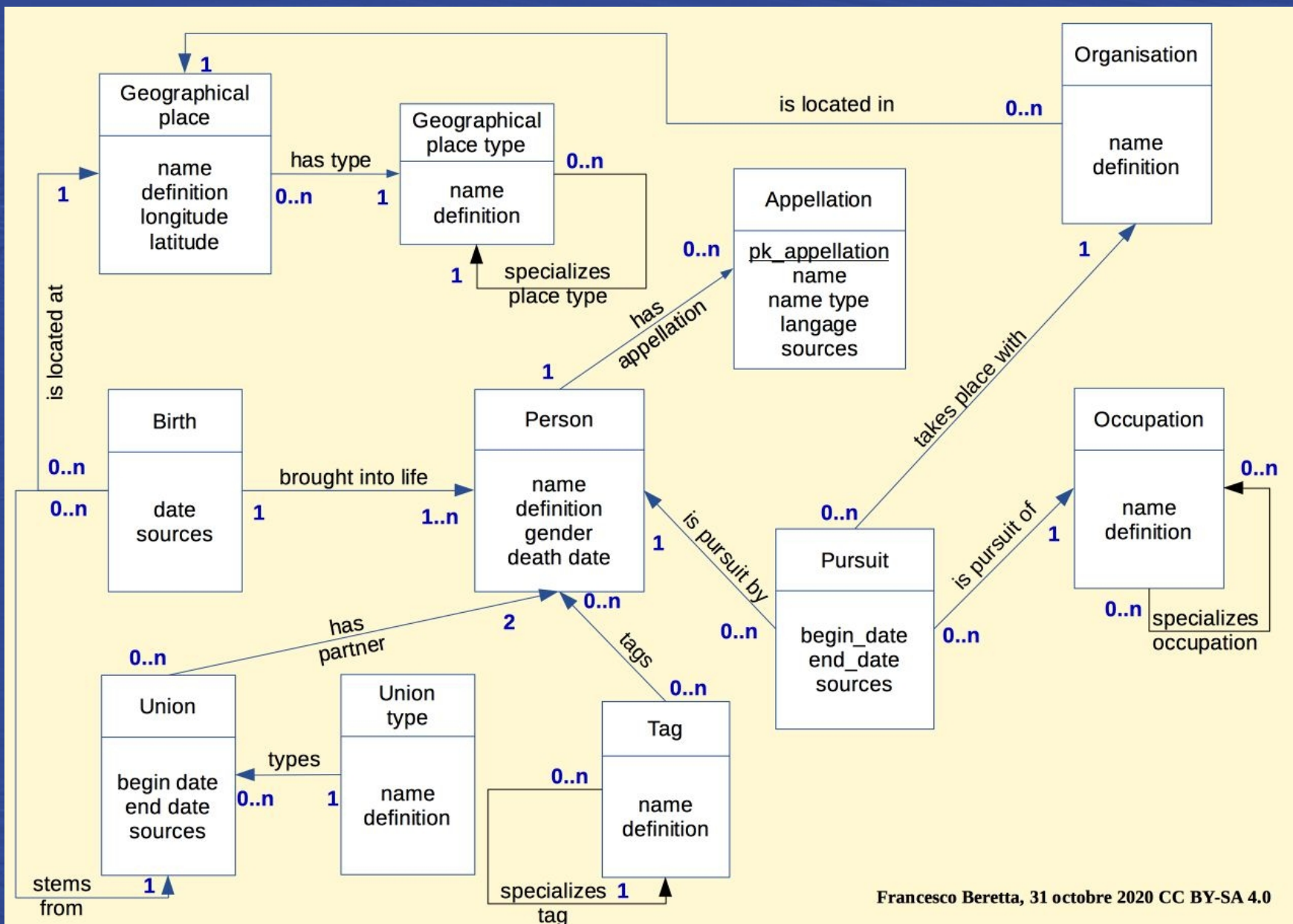
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Modèle conceptuel et base de données relationnelle



Francesco Beretta, 31 octobre 2020 CC BY-SA 4.0

“An ontology is
a formal explicit specification
of a shared conceptualization
of a domain of interest”

- « Formality – ... a knowledge representation language that is based on the grounds of **formal semantics**. »
- « Consensus – ... an agreement on a domain conceptualization among people in a community. »
- « Conceptuality – ... in terms of conceptual symbols that can be intuitively grasped by humans, as they correspond to the elements in their **mental models**. »
- « Domain Specificity – ... limited to knowledge about a particular **domain of interest**. »

[Domingue et al. 2011, p. 510-511]

OntoClean

N. Guarino/C. A. Welty, « An Overview of OntoClean », in Steffen Staab, ed., Handbook on ontologies, 2nd ed. Berlin: Springer, 2009.

Essence (as defined by rigid intensional properties)

- A 'student' : not a class but a time-indexed property of a person

Identity (criteria to clearly distinguish individuals)

- The identity of a building as individual does not depend on its changing use

Unity (parthood)

- A crowd of humans vs a group having a plan

Dependence (one instance implies the existence of another one)

- No human without a birth and no birth without a human

Modelling Best Practices

- Property inheritance, quantifiers, multiple statements on same properties
- Rich controlled vocabularies (researchers) but concise ontology (semantic engineers)

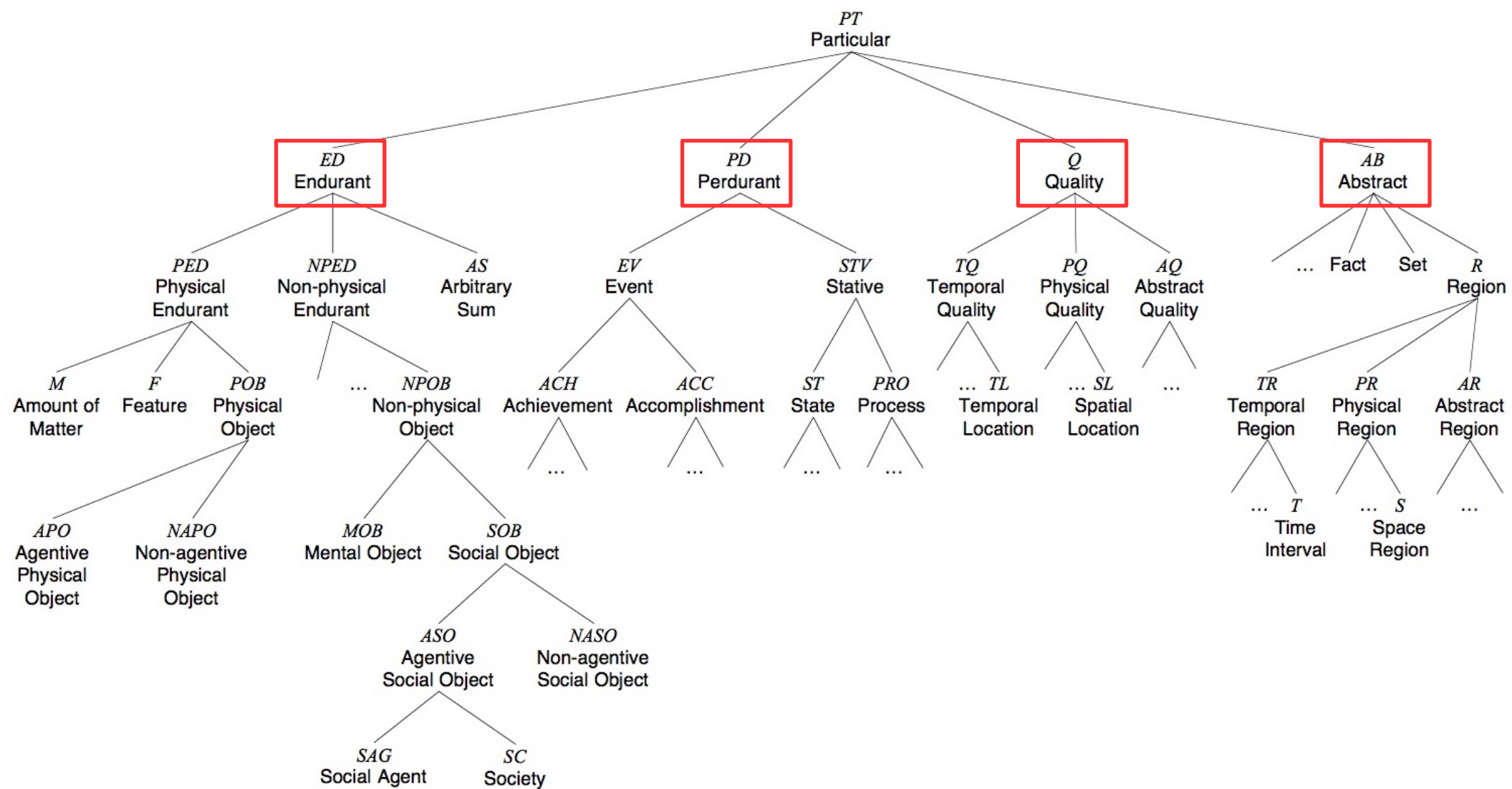
The issue of a suitable conceptualization :

Foundational ontologies
were developed to support the
verification and improvement
of the **conceptualization** of a **domain of discourse**.

« **Foundational ontologies** are not directly used for applications; rather, they provide **conceptual handles** to solve cases of misunderstandings due to the limitations of expressiveness of the application languages. [...]

DOLCE has remained fixed over the years fulfilling the purpose of top-level ontologies to provide a solid and stable basis for **modeling different domains**, in this way **ensuring interoperability of reference and domain ontologies** that use DOLCE. »

Borgo Stefano et al., « DOLCE: A descriptive ontology for linguistic and cognitive engineering¹ », Applied Ontology, 18.11.2021, pp. 1-25.



Descriptive Ontology for Linguistic and Cognitive Engineering (DOLCE) – a foundational ontology designed in 2002 in the context of the WonderWeb EU project, developed by Nicola Guarino and his associates at the Laboratory for Applied Ontology (LOA) – WonderWeb Deliverable D18, p.14

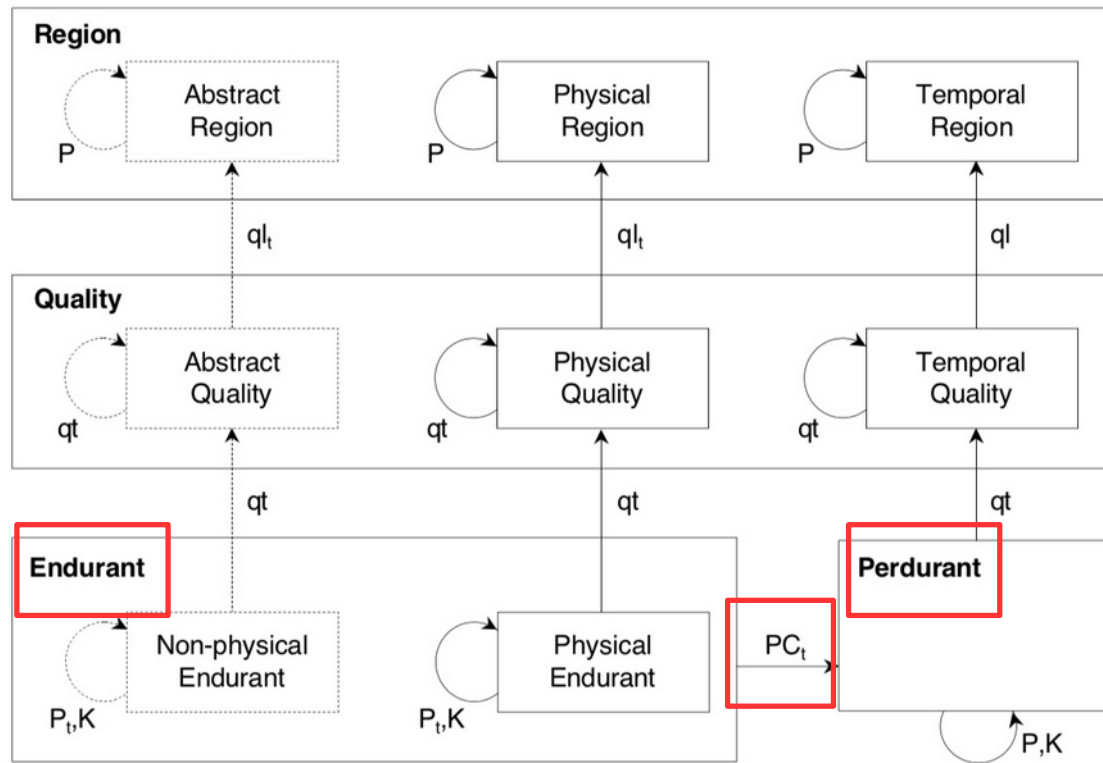


Figure 4: Primitive relations between basic categories (the dotted lines to the left indicate that we are less confident with what concerns non-physical endurants).

4.2.7 Participation

(Dd63) $PC_C(x, y) \triangleq \exists t (PRE(y, t)) \wedge \forall t (PRE(y, t) \rightarrow PC(x, y, t))$ (Const. Participation)

(Dd64) $PC_T(x, y, t) \triangleq PD(y) \wedge \forall z ((P(z, y) \wedge PRE(z, t)) \rightarrow PC(x, z, t))$
(Temporary Total Participation)

(Dd65) $PC_T(x, y) \triangleq \exists t (ql_T(t, y) \wedge PC_T(x, y, t))$ (Total Participation)

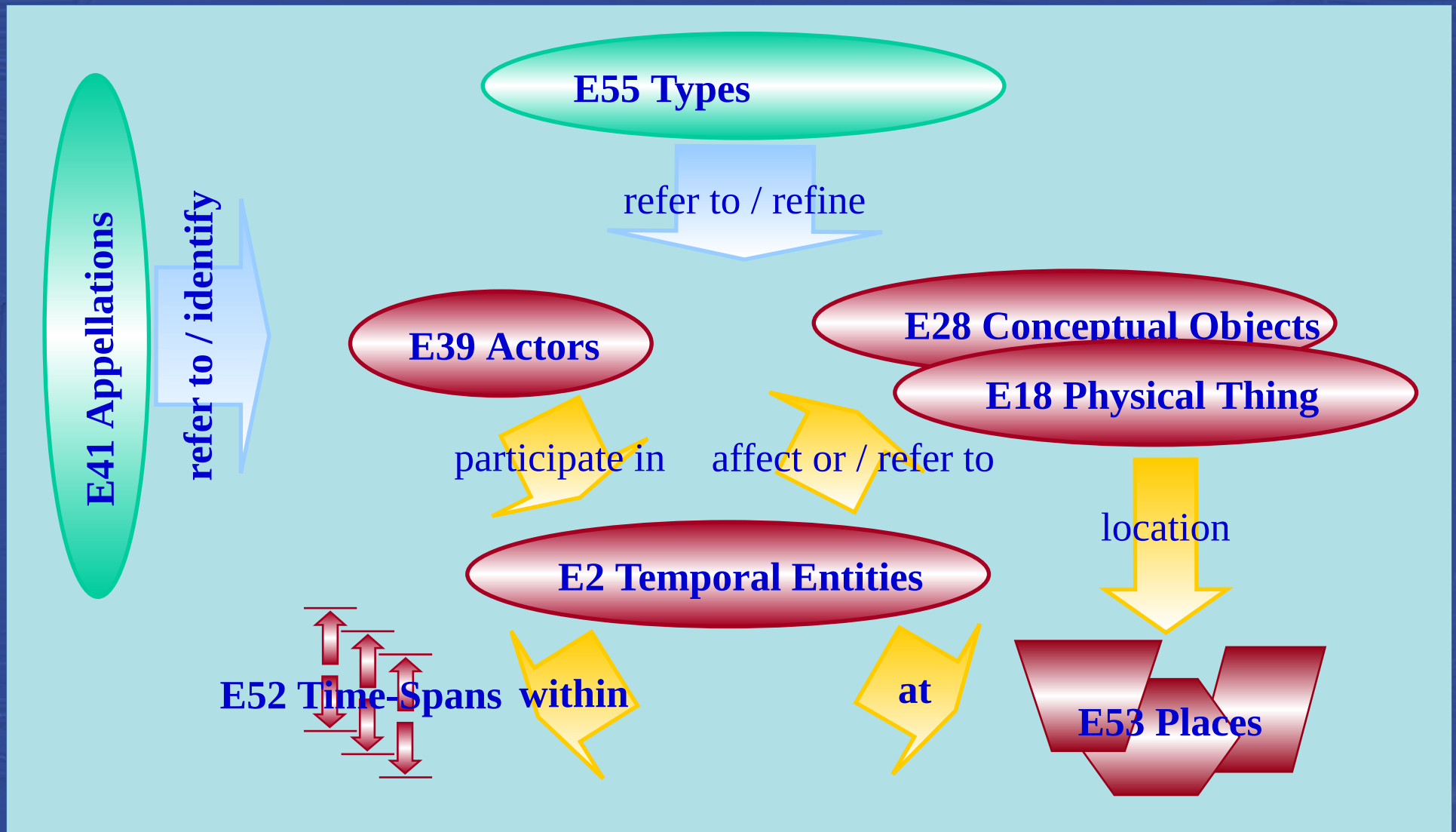
(Dd66) $mpc(x, y) \triangleq x = \sigma_t z (PC_T(z, y))$ (Maximal Participant)

(Dd67) $mppc(x, y) \triangleq x = \sigma_t z (PC_T(z, y) \wedge PED(z))$ (Maximal Physical Participant)

(Dd68) $lf(x, y) \triangleq x = \sigma z (PC_T(y, z))$ (Life)

The CIDOC CRM (ISO21127:2006)

A semantic framework that provides *interoperability* between different sources of **cultural heritage information**



Persistent items

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Person

Group

Geographical place

Conceptual object

Conceptual object

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Person

Person

Person

Person

Conceptual object

Conceptual object

Person

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Temporal entities

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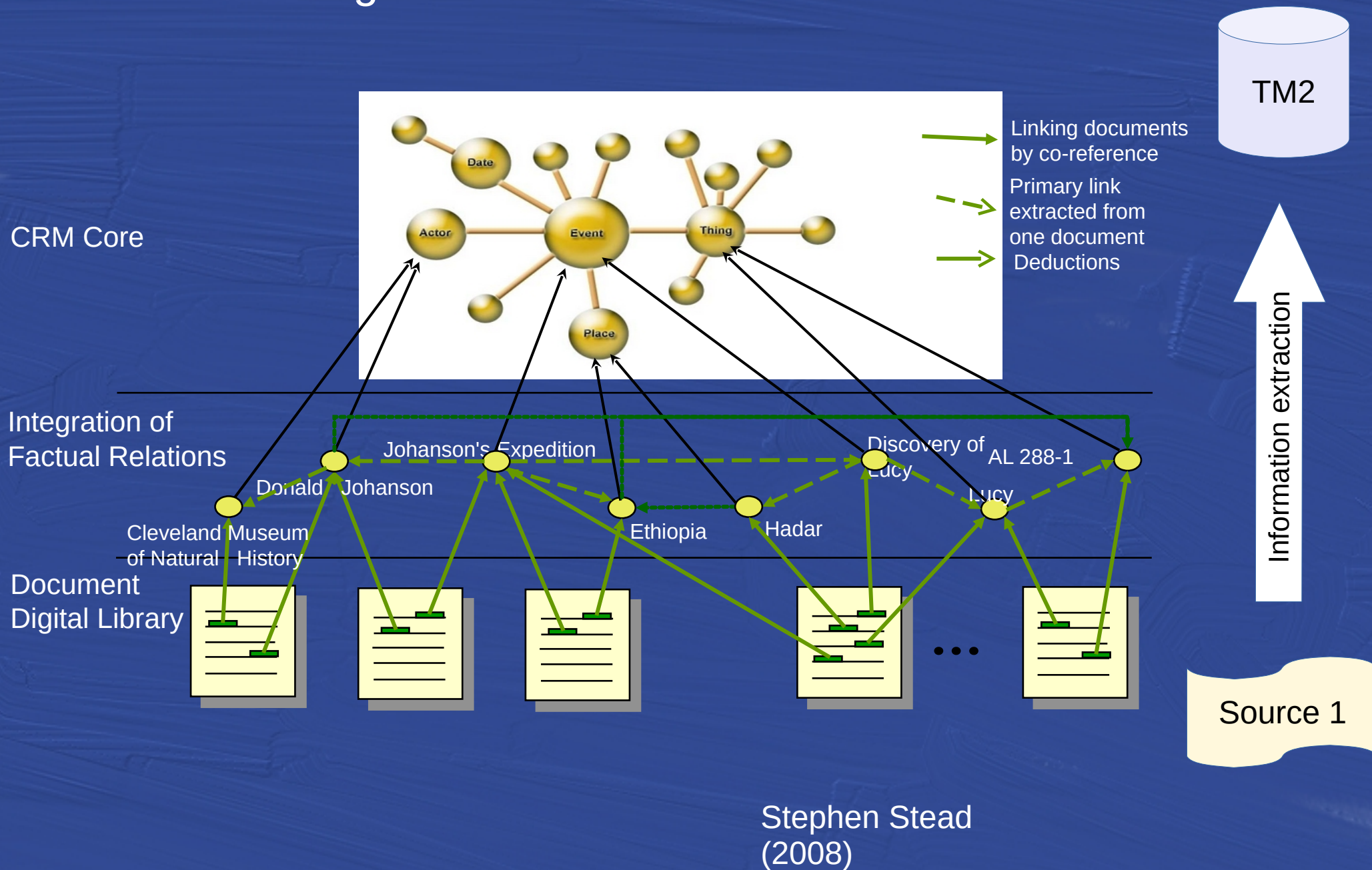
Manual encoding of the text or use of NLP techniques to identify information and produce structured data

<p corresp="Info29659"><name ref="Actr195">Johannes Kepler</rs>
was born on <date when="1571-12-27">December 27, 1571</date> in
<name ref="NaPI90073">Weil der Stadt</name></p>

```
{ "Tokens" : [  
  { "doc" : 1, "sentence" : 6, "token_id":1,  
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  { "doc" : 1, "sentence" : 6, "token_id":2,  
    "token" : was},  
  { "doc" : 1, "sentence" : 6, "token_id":3,  
    "token" : born},  
  { "doc" : 1, "sentence" : 6, "token_id":4,  
    "token" : on}, was  
  { "doc" : 1, "sentence" : 6, "token_id":5,  
    "token" : December},  
  { "doc" : 1, "sentence" : 6, "token_id":6,  
    "token" : 27},  
  { "doc" : 1, "sentence" : 6, "token_id":7,  
    "token" : 1571},  
  ... ]}
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Chunk [Tokens ids]	Entity	Value
1	Actr195	
7,5,6		1571-12-27

Integration of information extracted from documents using the CIDOC CRM and its extensions



Integration of information extracted from documents using the CIDOC CRM

